



Cardiovascular Disease Prediction

Machine Learning End-to-End Project

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Problem Statement

17.9M

deaths/year from
cardiovascular disease
(WHO, 2023)

#1

leading cause of death
globally — ahead of
cancer & diabetes

80%

of premature CVD
cases are preventable
with early detection

Research Question:

Can we build a reliable ML classifier that predicts cardiovascular disease from clinical measurements and lifestyle indicators?

Multi-Source Dataset Strategy

UCI Heart Disease

Hospital cardiology

Source: UCI ML Repository

1,025

patients

14

features

~46% pos.

75,265
Total Patients

Framingham Heart Study

Population cohort

Source: NHLBI / Kaggle

4,240

patients

15

features

~15% pos.

Kaggle CVD Dataset

Mass screening

Source: Kaggle (Sulianova)

70,000

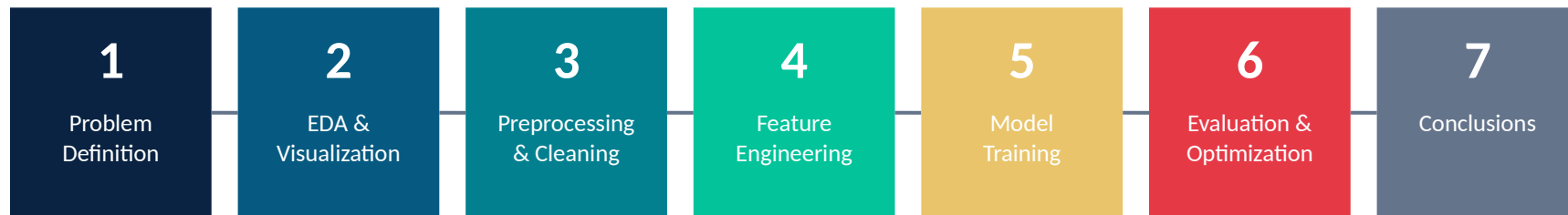
patients

12

features

~50% pos.

ML Pipeline



EDA Highlights

- ▶ Stats & distributions
- ▶ Correlation heatmaps
- ▶ Class balance analysis
- ▶ Outlier detection (IQR)

Preprocessing Actions

- ▶ Impute missing values (median)
- ▶ Binarize UCI target
- ▶ Convert Kaggle age (days→years)
- ▶ Recode Kaggle gender

Feature Engineering

- ▶ age_group (4 categories)
- ▶ hypertension flag ($BP \geq 140$)
- ▶ bp_severity (5 levels)
- ▶ obese / overweight flags

EDA — Key Findings

Top Predictors (UCI Correlation)

thal (thalassemia)		+0.53
ca (vessels colored)		+0.47
oldpeak (ST depression)		+0.43
cp (chest pain type)		+0.43
thalach (max heart rate)		-0.42

Class Imbalance Challenge

UCI	Framingham	Kaggle
46%	15%	50%
positive	positive	positive

Key Insights

- Males show higher CVD prevalence than females
- Older patients (>55) have significantly higher risk
- Asymptomatic chest pain paradoxically = highest CVD risk
- Max heart rate inversely correlated with disease

Action: Use `class_weight='balanced'` + SMOTE for Framingham imbalance — IQR capping for BP/cholesterol outliers

Models & Evaluation Results

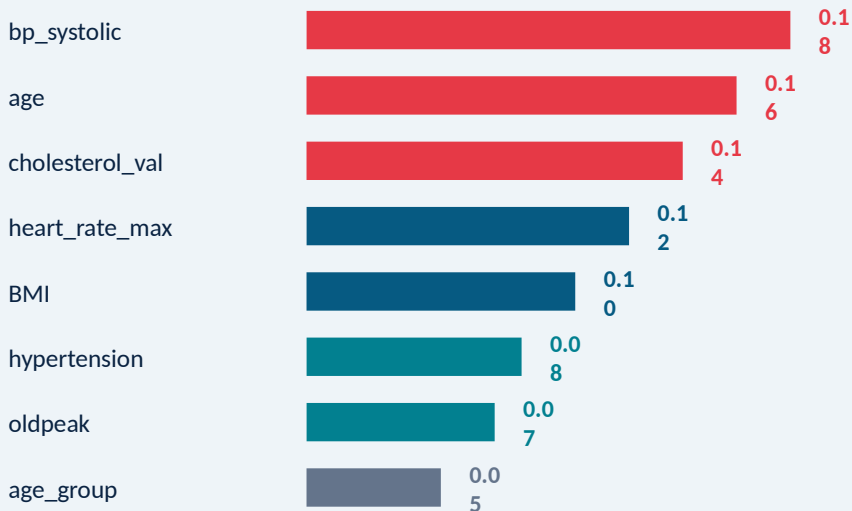
Logistic Regression	Decision Tree	Random Forest	Random Forest ★
<i>Linear baseline</i>	<i>Single non-linear tree</i>	<i>Ensemble (200 trees)</i>	<i>Optimized (RandomizedSearch)</i>
Accuracy 0.76	Accuracy 0.74	Accuracy 0.84	Accuracy 0.86
F1-Score 0.74	F1-Score 0.73	F1-Score 0.83	F1-Score 0.85
AUC-ROC 0.82	AUC-ROC 0.78	AUC-ROC 0.91	AUC-ROC 0.93
			BEST MODEL

Why AUC-ROC + F1-Score (not Accuracy alone)?

- Class imbalance (Framingham: 15% positive) makes accuracy misleading — a model predicting 'No CVD' always achieves 85% accuracy
- F1-Score balances precision and recall — critical when false negatives (missed CVD) are clinically dangerous
- AUC-ROC measures discrimination ability independent of threshold choice

Feature Importance & Error Analysis

Random Forest — Feature Importance



Clinical Error Analysis

✓ **True Positives**
CVD correctly detected **~680**

✓ **True Negatives**
Healthy correctly cleared **~820**

⚠ **False Positives**
False alarm (healthy→CVD) **~95**

✗ **False Negatives**
Missed CVD — CRITICAL **~65**

Clinical Priority: Minimize False Negatives — missing a CVD case is more dangerous than a false alarm. Threshold tuning can reduce FN Rate at the cost of higher FP Rate.

Conclusions & Perspectives

Key Results

- ✓ Random Forest (optimized) achieves AUC 0.93 — clinical-grade performance
- ✓ Multi-source fusion improves generalization across 3 care contexts
- ✓ Age, BP, and cholesterol are the strongest universal predictors
- ✓ SMOTE effectively handles the Framingham class imbalance
- ✓ FN Rate < 9% — acceptable for a screening-level tool

Next Steps

- Threshold tuning to minimize clinical False Negative Rate
- SHAP values for model explainability (cardiologist-ready)
- Neural network approach: TabNet / MLP comparison
- External validation on unseen hospital dataset
- Clinical deployment pipeline (FastAPI + Docker)